

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10: 3.2 CURRENT ELECTRICITY – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the Huruma Estate phenomenon. Use what you learned in Lessons 1 to 12 to answer every section in the space under each prompt. For each answer, state a claim, give evidence from our investigations, and explain your reasoning using ideas such as Ohm's Law, EMF, internal resistance, series and parallel rules, power and energy. Include units in every calculation and show your working. Write in full sentences.

Section 1: Dim lights and a slow kettle after the blackout

After a Kenya Power blackout in Huruma Estate, Nairobi, the lights glowed dimly and the kettle took longer to boil when electricity was first restored. A few minutes later everything worked normally. Explain what happened. In your answer: (a) say what you measure with a voltmeter on the supply terminals when current is flowing compared with when no current flows; (b) use the idea of internal resistance and the equation for terminal voltage; (c) link the change in voltage to the power delivered to the bulbs and kettle using $P = V^2/R$.

Write your answer here:

Section 2: The bulb on the long, thin extension cord

A neighbour in Huruma Estate has a bulb on a long, thin extension cord. It is always dimmer than a bulb plugged directly into the wall socket, even when power is fully restored. In the Huruma Extension Cord Circuit from Lesson 7, the data were:

Part of circuit	Voltage (V)
Supply	240
Across the 6 Ω extension cord	31.3
Across the parallel appliances (bulb in Socket A and charger in Socket B)	208.7

(a) Explain why the bulb is dimmer, using the data. (b) Describe how the cord's length and cross-sectional area affect its resistance, and how this links to $\rho = RA/L$. (c) Calculate the current flowing through the cord. (d) Explain why this is a problem for energy delivery and safety.

Write your answer here:

Section 3: Why homes are wired in parallel

Explain why all the appliances in a Kenyan home are connected in parallel to the 240 V supply rather than in series. Use what you observed in Lessons 5 and 6 (Christmas-style bulbs in series and the parallel circuits you built). Mention voltage, current and what happens when one appliance is switched off or removed.

Write your answer here:

Section 4: Power, energy and cost at home

A 3 kW electric jiko is used in a Nairobi home on a 240 V supply. Kenya Power charges about Ksh 12.00 per unit (1 kWh). (a) Calculate the current the jiko draws. (b) Calculate how long it takes to use 1 kWh and the cost of running it for that time. (c) Explain why a kettle that runs on lower terminal voltage during the Huruma power restoration takes longer to boil water, and say whether the energy needed to boil the same water changes.

Write your answer here:

Section 5: Designing a safe, efficient circuit for the Wanjiku Family Home

The Wanjiku family in Githurai, Nairobi, has a 15 A mains fuse on 240 V. They want to run the 3 kW electric jiko and a second 3 kW appliance at the same time on this fuse. (a) Calculate the total current if both run together and decide whether the 15 A fuse is safe. (b) Explain what the fuse does and what could happen if a much bigger fuse or a thin extension cord were used instead. (c) Give three design rules, with reasons, for wiring the home so that every appliance gets the right amount of energy safely and efficiently. Answer the driving question: why do devices sometimes get less energy than expected, and how do we design circuits to prevent this?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim, evidence and reasoning	Gives a clear claim for each phenomenon, supports it with specific data from our investigations (for example 31.3 V and 208.7 V), and links evidence to the claim with correct scientific reasoning.	Gives claims and some evidence from the lessons, but reasoning is partly missing or does not fully link the evidence to the claim.	Gives statements with little or no evidence, or evidence that does not match the claim.
Use of circuit ideas ($V = IR$, internal resistance, series and parallel rules)	Uses $V = IR$, $V = E - Ir$, and the series and parallel rules correctly and consistently to explain both the post-blackout dimming and the extension cord.	Uses most of the ideas correctly, with a few minor errors or missing links.	Uses ideas incorrectly or confuses series with parallel, or EMF with terminal voltage.
Calculations of power, energy, current and cost	Calculations are correct with formula, substitution, working and units (A, W, kWh, Ksh).	Method is correct, but there are small arithmetic or unit errors.	Method is incorrect or incomplete, or units are missing.
Safe and efficient design decisions	Justifies parallel wiring, thick cables, fuse choice and earthing, and links each to the driving question about energy delivery and safety.	Gives design rules with partial justification.	Gives design rules without reasons, or rules that are unsafe.

Communication and Kenyan context	Writing is clear and well organised, uses correct terms, and refers to the Huruma Estate and Kenya Power contexts.	Writing is mostly clear, with some terms misused or limited reference to context.	Writing is hard to follow, with little use of scientific terms or context.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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